Wood Planter Box

Name: _____
Date: _____

Description:

Students will complete a wood working project by building a wood planter box. The planter box will be big enough to hold 4 – 4" square pots or can be filled with potting soil. Great project for Christmas or Mother's Day.

Materials:

1" X 6" X 6' Redwood
5d galvanized box nails
3d galvanized box nails (few)
Wood Glue
Varnish/stain
Sand Paper

Tools:
Table Saw
Power Miter Saw
Drill Press
1" forstner bit
Hammer
Tape Measurer/Ruler/Combination square

Procedure:

1.) Obtain a piece of redwood that is 1" X 6" X 6'
2.) Determine the best use of the lumber. Where are the knots? Determine the layout before you begin to cut.
3.) Crosscut the board at approximately 24" for the bottom.
4.) Adjust the table saw rip fence so it measures 5" from the fence to the inside of the saw blade
5.) Rip (cut) the long piece of redwood at 5".
6.) From this piece cross cut all the 5" wide pieces.
7.) Set the table saw to rip at 4 ¼". Rip your short board. Save the scrap for the feet.
8.) Crosscut the 4 ¼" board at 17".
9.) Crosscut the scrap for the feet.
10.) All of the pieces have now been cut. Put your project together without glue and nails. Does it fit? If so continue.
11.) Layout the holes in the bottom. Drill 4 holes using a Forstner bit evenly spaced under each pot.
12.) Nail the ends to the bottom using glue and 5d nails.
13.) Nail the sides to the bottom and ends using glue and nails. Be sure to properly align the sides (see plan) and insure the ends are square.
14.) Turn the planter upside down and install the feet as shown the plan. Use 3 or 4d nails and glue. Brittle wood may need to be drilled.
15.) Sand your project down to a smooth finish, DO NOT sand the nails.
16.) Stain or paint the project for completion

Cutting List:

Quantity	Size		Material
2	3/4” X 5 ” X 20”	Sides	Redwood
1	3/4” X 4 ¼” X 17”	Bottom	Redwood
2	3/4” x 5” X 4 ¼ ”	Ends	Redwood
2	3/4”X 5 ¾” X 1”	Feet	Redwood

Notes:

_____

_____

_____

_____

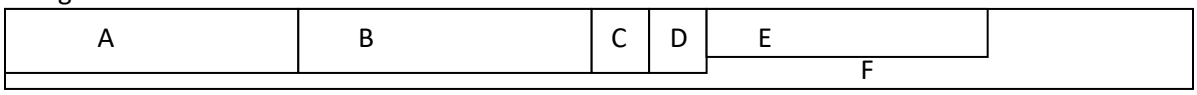
_____

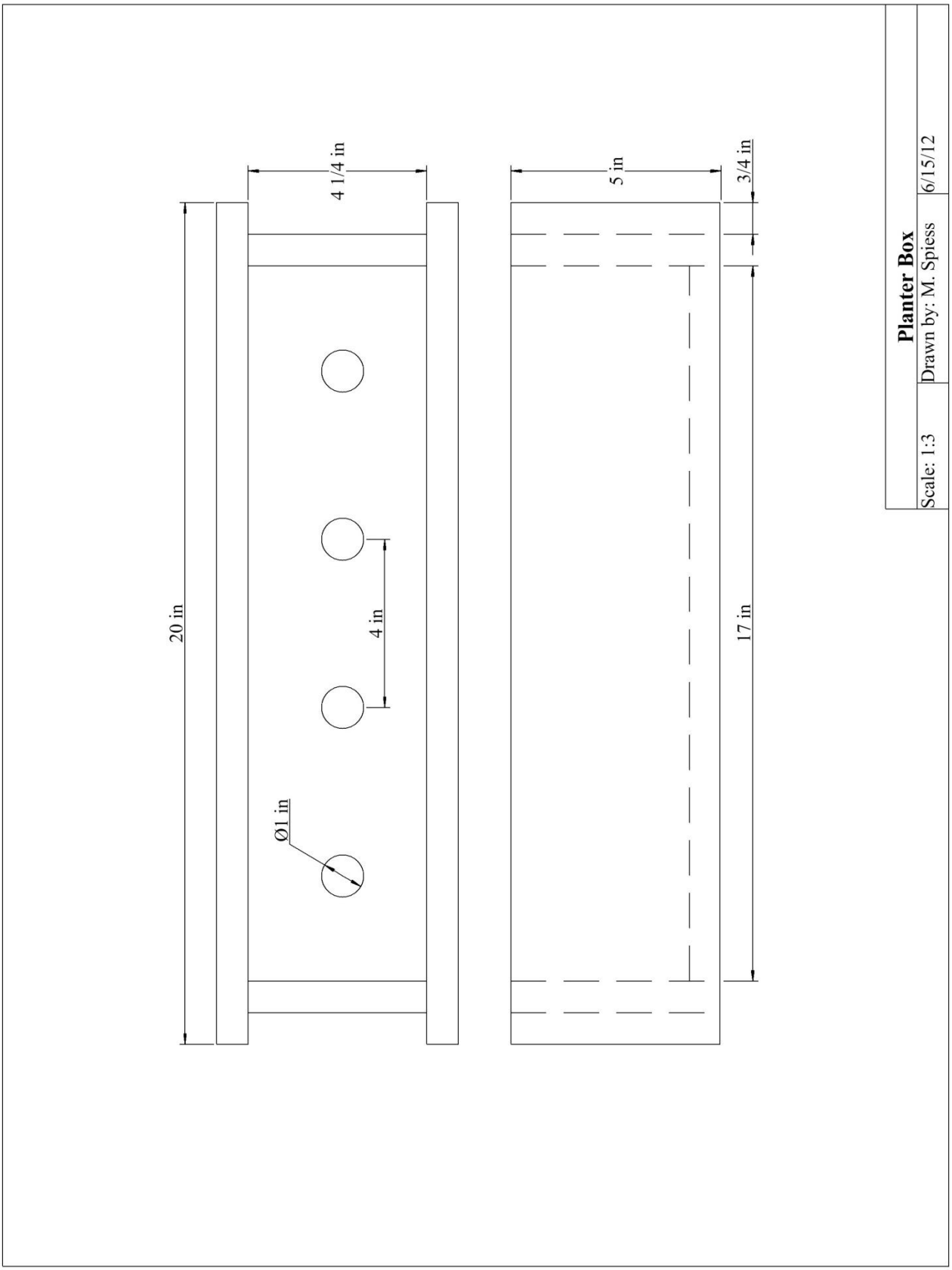
_____

Photo/Drawing:

Board Layout

A=Side Piece
B=Side Piece
C=End Piece
D=End Piece
E=Bottom Piece
F=Legs




20 in
4 1/4 in
4 in
Ø1 in
5 in
3/4 in
17 in
Planter Box
Scale: 1:3
Drawn by: M. Spiess
6/15/12

Wood Planter Box Worksheet

Name: _____
Date: _____

1. What are the safety precautions when using the table saw?

2. When using a table saw why must two people help rip a large piece of wood?

3. When using the table saw why must you always measure from the fence to the inside of the saw blade?

4. When using the table saw why must the top of the saw blade be 1/8” above the top of the wood?

5. What are the safety precautions when using the power miter saw?

6. What type of drill bit is used and why?

Grading Rubric:

CRITERIA	POSSIBLE	SCORE
Length	5	
Width	5	
Holes are properly drilled-no chips, slivers, etc.	5	
Holes are lined up straight and evenly spaced	5	
Planter box is square and joints fit well.	5	
Workmanship (sanding, nails, cut quality, etc)	5	
Total	**30**	

Wood Planter Box Teachers Notes:

Agricultural Standards Met:

6.0 Health and Safety. Students understand health and safety policies, procedures, regulations, and practices, including the use of equipment and handling of hazardous materials:

- 6.1 Know policies, procedures, and regulations regarding health and safety in the workplace, including employers' and employees' responsibilities.
- 6.2 Understand critical elements of health and safety practices related to storing, cleaning, and maintaining tools, equipment, and supplies.
- 6.4 Maintain safe and healthful working conditions.
- 6.5 Use tools and machines safely and appropriately.
- 6.6 Know how to both prevent and respond to accidents in the agricultural industry.

B1.0 Students understand personal and group safety:

- B1.1 Practice the rules for personal and group safety while working in an agricultural mechanics environment.
- B1.2 Know the relationship between accepted shop management procedures and a safe working environment.
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B2.0 Students understand the principles of basic woodworking:

- B2.1 Know how to identify common wood products, lumber types, and sizes.
- B2.2 Know how to calculate board feet, lumber volume, and square feet.
- B2.3 Know how to identify, select, and implement basic fastening systems.
- B2.4 Complete a woodworking project, including interpreting a plan, developing a bill of materials and cutting list, selecting materials, shaping, jointing, and finishing.

Objectives:

By properly completing this project, students will be able to:

- Read a plan to and layout dimensions.
- Students will identify proper wood type used for the project
- Students will properly layout project design
- Students will safely use a table saw
- Students will safely cut wood using the power miter saw
- Students will use a drill press to successfully drill drainage holes
- Students will learn the proper way to glue and assemble the project
- Students will learn how to sand and stain their planter box

Alternative Tools/Methods/Materials:

- Many simple variations are possible. Joints may use dado and rabbet cuts if a dado blade is available. Dado and rabbet cuts can also be made with a router or shaper (see alternate plans)
- Instead of using nails you could have used #6 x 1 ½ ” deck screws.
- Cedar or pine instead of redwood. Pine would need to be painted.

Safety Review:

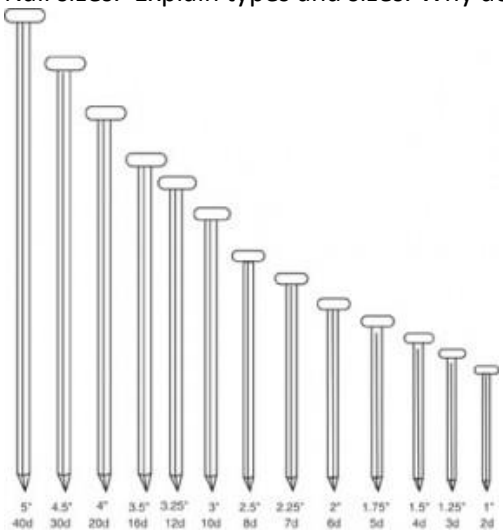
- Table Saw
- Power Miter Saw
- Drill Press

Project Time:

Demonstration:	1.5 hours
Build:	1.5 hours

Demonstration Notes:

1.) Demonstrate how to select lumber, cutting around knots. Project allows for some waste. You may wish to simply rip all of the boards. Cut three 5” boards for every 4 ¼” board.
2.) Be sure to address safety tips as we work on each machine.
3.) Demonstrate mark-cut-mark-cut technique used for cross cutting so pieces are correct length.
4.) Demonstrate hole layout using a combination or try square.
5.) Drill holes before legs are glued and nailed on to the bottom piece.
6.) Make sure that end pieces are placed at the 2” mark (from the end of the board). Make sure that they are placed at the 2” mark on the bottom piece (on both ends) and that they match up with the side piece 2” mark.
7.) Make drainage holes evenly spaced and in a straight line.
8.) Keep scrap pieces separate to avoid confusion.
9.) Nail sizes. Explain types and sizes. Why don’t we use 5d on feet?



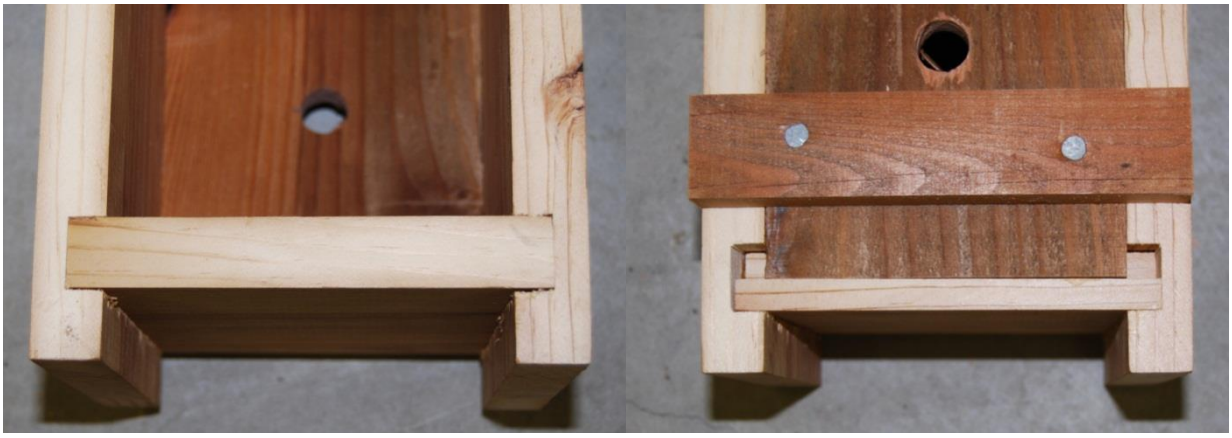
Bill of Materials:

Projects:	24					
Size	Description	Units	Qty/Project	Cost/Unit	Order	Amount
1"X 6" x 12'	Redwood	each	0.5	$9.00	12	$ 108.00
5d	Galv common nails	Pound	0.05	$ 2.00	2	$ 4.00
						$ -
						$ -
						$ -
					TOTAL	$ 112.00
					Each	$ 4.67

Project from: Mike Spiess, Mike Mederos.

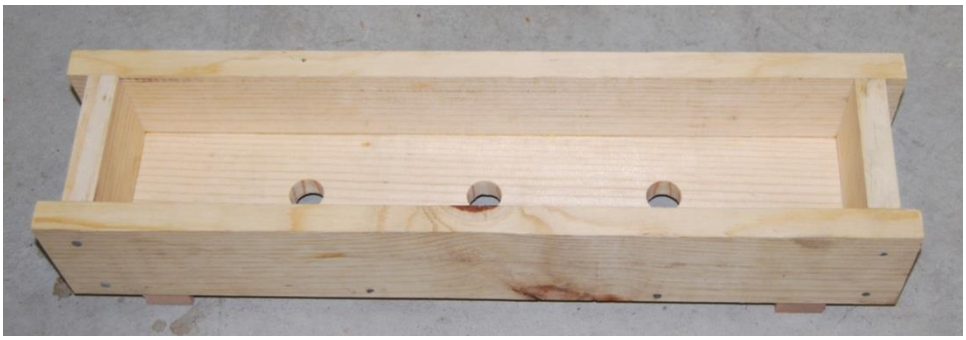
Alternate Projects:

There are many variations on this project. Here are a few….



A version of the project with dado and rabbet cuts. Good project to introduce more complex wood joints. Requires a table saw with a dado blade.

A version with a decorative back. Can be wall mounted. Use a paper template (another layout skill) to make the back shape. Cut curves with a ¼" bandsaw.



A simple version made from 1x4. No tablesaw needed. Use 3" pots or fill with soil.

A hanging version made from 1x4. Use this 3” pots. Cut ends on bandsaw with ¼” blade. Note use of finish nails. Another way to introduce a different type of fastener.

Simple cedar box built around a commercial tray (Home Depot). Good for indoor use.